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Q7. (a) What is recombinant DNA technology? How it is done? Discuss its applications 15

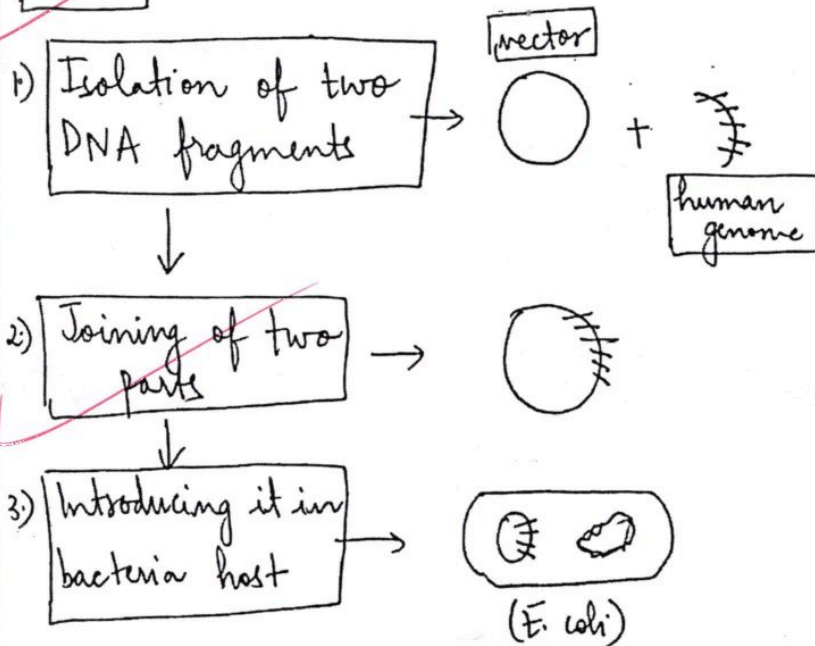
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~~Recombinant DNA~~ Recombinant DNA (r-DNA)

technology is a process in which two DNA parts of different species are joined.

eg: joining of human DNA vector with plasmid of bacteria (*E. coli*).

Process



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4) Cloning of DNA → 

↓
5) Segregation for use

Applications

- 1) gene mapping → knowing structure of a particular gene.
- 2) gene therapy particularly in gene replacement
- 3) Synthesis of vaccines like BCG.
- 4) National ~~Dairy~~ Dairy Institute has synthesized growth hormone and insulin which has huge applications in dairy sector.
- 5) Application in agricultural sector. eg: production of hybrid seeds.

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6) ~~Ex~~ It can be used to produce variety of fishes thus helping in blue revolution.

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Challenges

- 1) Ethical issues: related to genetic information.
- 2) Any harmful organism can be produced which can negatively affect ~~pro~~ biodiversity.
- 3) Such introduction & experiments on bacteria can make it ~~antibiotic~~ antibiotic resistant.

Despite all this, r-DNA has immense application like reverse genetics and genetically modified organisms. It is also being used in ~~reverse~~ developing vaccines of COVID-19.